

5.12.2018 (E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov-Dec 2018

Duration: 3 hrs
Semester: VI

Max. Marks: 20

Class: TY Comp

Name of the Course: System Programming and Compiler Construction (SPCC)

Branch: Comp

Course Code: UCEC 602

Instructions:

- (1) All Questions are Compulsory**
- (2) Draw neat diagrams**
- (3) Assume suitable data if necessary**

Q 5. A) Describe in brief .

- a> What is LR(k) parser. What is the difference between LR(0) and LR(1) parser. 2M
- b> What is the meaning of first and follow. Write rules of First . 2M
- c> What is the need of intermediate codes. 2M
- d> What is the meaning of peephole optimization. 2M
- e> What is forward reference problem ? 2M

Q 5 B) Answer the following questions with justification. Each Q carry 2 M.

1) Relocation bits used by relocating loader are specified by
(A) Relocating loader itself (B) Assembler or Translator
(C) Macro processor (D) Both (A) and (B)

2) Relocatable programs
(A) cannot be used with fixed partitions
(B) can be loaded almost anywhere in memory
(C) do not need a linker
(D) can be loaded only at one specific location

3) In a two pass assembler the object code generation is done during the

- (A) First pass
- (B) Zeroeth pass
- (C) Second pass
- (D) Not done by assembler.

4) Which of the following is not a type of assembler ?

- (A) one pass
- (B) two pass
- (C) three pass
- (D) load and go

5) Which of the following system program foregoes the production of object code to generate absolute machine code and load it into the physical main storage location from which it will be executed immediately upon completion of the assembly ?

- (A) Macro processor
- (B) Load and Go assembler
- (C) Two pass assembler
- (D) Compiler

03-12-18(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Examinations
Nov – Dec 2018

Max. Marks:100

Class: Third Year B. Tech.

Name of the Course: Artificial Intelligence

Course Code: UCEC 601

Duration: 3 hrs.

Semester: VI

Branch: Computer

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks
Q 1 (a)	What is PEAS descriptor? Explain PEAS descriptor for Self driving car	10
Q 1 (b)	Explain Utility based agent with architecture.	10
Q2 (a)	Compare uninformed search strategies with respect to time complexity, space complexity, optimality and completeness.	10
	OR	
Q2 (a)	Apply state space search method for Water Jug problem.	10
Q2 (b)	Give the problems associated with Hill Climbing algo. Suggest the solutions.	10
Q3 (a)	Explain A* search algorithm with example	10
	OR	
Q3 (a)	Explain Best first search algorithm with example.	10
Q3 (b)	Apply a partial-order plan for putting on shoes and socks, and analyze their six corresponding linearizations into total-order plans.	10
Q4 (a)	Explain forward chaining and backward chaining with example.	10
	OR	
Q4 (a)	Write a program in PROLOG to solve the Monkey Banana problem. Imagine a room containing a monkey, chair and some bananas. That have been hanged from the centre of ceiling. If the monkey is clever enough he can reach the bananas by placing the chair directly below the bananas and climb on the chair .The problem is to prove the monkey can reach the bananas. The monkey wants it, but cannot jump high enough from the floor. At the window of the room there is a box that the monkey can use. The monkey can perform the following actions:- 1) Walk on the floor. 2) Climb the box. 3) Push the box around (if it is beside the box). 4) Grasp the banana if it is standing on the box directly under the banana.	10
Q4 (b)	Represent following sentences in first order logic. 1. Every Austinite who is not conservative loves some armadillo.	10

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	2. Anyone who wears maroon-and-white shirts is an Aggie. 3. Every Aggie loves every dog. 4. Everyone in the world is loved by at least one person. 5. Either John or Peter will win, but not both.	
Q5 (a)	Explain Expert system development steps with suitable example.	10
Q5 (b)	Explain Learning agent with diagram.	10
	OR	
Q5 (b)	Write a short note on Bayesian networks	10

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K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Examinations

Nov – Dec 2018

Max. Marks: 100

Class: ty comp.

Name of the Course: System Programming and Compiler Construction

Branch: Computer

Duration: 3 hours

Semester: VI

Course Code: UCEC602

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks																											
Q1(a)	Explain the different functions of loader in brief.	05																											
Q1(b)	Compare compiler and interpreter.	05																											
Q1(c)	Explain left recursion with an example.	05																											
Q1(d)	What is activation record? Draw the diagram of general Activation record and explain the purpose of different fields of an activation record	05																											
Q2(a)	Differentiate between Linkage editor and direct linking loader	05																											
Q2(b)	What is Input Buffering? State its advantages	05																											
Q2(c)	Explain the different phases of compiler. Illustrate the output after each phase for the following statement: if(a>5) a=8; else a=3;	10																											
	OR																												
Q2(c)	Explain the runtime storage organization in detail.	10																											
Q3(a)	Explain different ways of parameter passing in macros.	05																											
Q3(b)	For the following grammar construct LL(1) parser table S->F S->(S-F) F->a Parse the String (a-a). Show the contents of stack and input buffer and action taken after each step.	10																											
	OR																												
Q3(b)	For following code what will be output generated by Pass-I and Pass-II for two pass assembler. Explain with database.	10																											
	<table border="1"> <tr> <td>ABC</td><td>Start</td><td>0</td></tr> <tr> <td></td><td>USING</td><td>*,15</td></tr> <tr> <td></td><td>L</td><td>1,FIVE</td></tr> <tr> <td></td><td>A</td><td>1,FOUR</td></tr> <tr> <td></td><td>ST</td><td>1,TEMP</td></tr> <tr> <td>FOUR</td><td>DC</td><td>F'4'</td></tr> <tr> <td>FIVE</td><td>DC</td><td>F'5'</td></tr> <tr> <td>TEMP</td><td>DS</td><td>1F</td></tr> <tr> <td></td><td>END</td><td></td></tr> </table>	ABC	Start	0		USING	*,15		L	1,FIVE		A	1,FOUR		ST	1,TEMP	FOUR	DC	F'4'	FIVE	DC	F'5'	TEMP	DS	1F		END		
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FIVE	DC	F'5'																											
TEMP	DS	1F																											
	END																												
Q3(c)	Differentiate between Top down and bottom up parser.	05																											

Q4(a)	Explain synthesized and inherited attributes used in Syntax Directed Definition	05																														
Q4(a)	OR Design Shift reduce parser for the following:- $E \rightarrow E + n \mid n$	05																														
Q4(b)	Mention the issues in the design of Code generator.	10																														
Q4(b)	OR Explain the different error recovery techniques used in a compiler.	10																														
Q4(c)	Differentiate between Compile and go loader, Absolute Loader and Relocatable loading Scheme	05																														
Q5(a)	Write short notes on: (any two) (a) Lex (b) Yacc (c) Basic Block and Flow graph	10																														
Q5(b)	Consider the following statements <table border="1"> <tr><td></td><td>MACRO</td><td></td></tr> <tr><td></td><td>INCR</td><td>&ARG1,&ARG2,&ARG3,&LAB</td></tr> <tr><td>&LAB</td><td>A</td><td>1,&ARG1</td></tr> <tr><td></td><td>A</td><td>2,&ARG2</td></tr> <tr><td></td><td>A</td><td>3,&ARG3</td></tr> <tr><td></td><td>MEND</td><td></td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td>INCR</td><td>DATA1,DATA2,DATA3,LOOP1</td></tr> <tr><td></td><td></td><td></td></tr> <tr><td></td><td>INCR</td><td>DATA3,DATA2,DATA1,LOOP2</td></tr> </table> Show how entries in MNT,MDT and ALA at end of pass I OR Explain Two Pass Macroprocessor with flowchart and Databases		MACRO			INCR	&ARG1,&ARG2,&ARG3,&LAB	&LAB	A	1,&ARG1		A	2,&ARG2		A	3,&ARG3		MEND						INCR	DATA1,DATA2,DATA3,LOOP1					INCR	DATA3,DATA2,DATA1,LOOP2	10
	MACRO																															
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Q5(b)		10																														

Date: 5/12/18

course: System programming & compiler Construction

course code: VCE1602

Question changed for specially abled students

Q.5b) what is need of macroprocessor
what are the different tables used in
pass-I & pass-II.

- OR

Q.5b) what is meaning of nested macro
call. Explain with example

Q.3b) what are the different tables
used in design of pass-I & pass-II
assembler

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Examinations
Nov – Dec 2018

Max. Marks:100

Class: THIRD

Name of the Course: Software Engineering

Course Code:UCEI602

Duration: 3 Hours

Semester: VI

Branch: Computer

IDC

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks
Q1 (a)	What do you mean by Software Engineering? Explain Capability Maturity Model (CMM) in details.	10
Q1 (b)	Explain the incremental and prototype model with merits and demerits.	10
	OR	
	What do you mean by requirements? Explain Functional and Non-Functional requirements in details with an example.	
Q2	Develop a Software Requirement Specification (SRS) for developing software for hospital management system. Create an SRS that contains the following. a) Objective and Scope b) Product perspective c) Functional requirements (at least 3) d) Non-functional requirements	20
Q3 (a)	What are the different categories of risks? Explain the process of risk projection.	10
Q3 (b)	Explain the various fundamental software design concepts.	10
	OR	
	What is user interface design process? explain with one example.	
Q4 (a)	What is coupling and cohesion? Explain different forms of coupling and cohesion.	10
Q4 (b)	Explain and compare Formal Technical Reviews and walkthrough.	10
	OR	
	What are the objectives of testing? Explain black box testing and integration testing.	
Q5 (a)	What are the different types of maintenance and also explain steps for creating maintenance log?	10
Q5 (b)	Write short notes on (any two). i. Agile Methodology. ii. Version control and change control activities in Software Configuration Management iii. COCOMO Model. iv. Pattern Based Software engineering	10

7.12.2018 (E)

K. J. Somaiya College of Engineering, Mumbai-77
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End Semester Examinations
Nov - Dec - 2018

Max. Marks: 100

Class: T.Y.B.Tech

Name of the Course: Digital Signal & Image Processing

Course Code: UCEC603

Duration: 3Hrs

Semester: VI

Branch: Computer

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks
Q 1	Solve any Four: i) Explain image fidelity criterion. ii) Convolution in spatial domain is multiplication in Fourier/frequency domain? Justify. iii) Find energy and power of unit step sequence. iv) Write $8 * 8$ Hadamard transform matrix and draw its signal flow graph. v) Explain gray level slicing image enhancement technique. vi) Explain sampling and quantization rate for digital image.	20
Q 2 (a)	Find linear and circular convolution of following $x(n) = \{1, 1, 0, 1, 1\}$ and $h(n) = \{1, -2, -3, 4\}$	10
Q2 (b)	Define discrete time signals and also give the classification of discrete time signals with suitable example. OR For a given discrete time system $y(n) = x(n) \sin(\omega n)$ Check whether the given system is: i) Static or dynamic ii) Linear or nonlinear iii) Shift invariant or variant iv) Causal or noncausal v) Stable or unstable	10
Q3 (a)	Give masks for the following operators and explain its usefulness in image processing. i) Sobel operator ii) Prewitt operator.	10

Q3 (b)	Explain DCT and its properties. Find the DCT for the following image: <table><tr><td>2</td><td>4</td><td>4</td><td>2</td></tr><tr><td>4</td><td>6</td><td>8</td><td>3</td></tr><tr><td>2</td><td>8</td><td>10</td><td>4</td></tr><tr><td>3</td><td>8</td><td>6</td><td>2</td></tr></table> OR Find 8 point DFT for given sequence {1, 1, 2, 0, 1, 2, 0, 1} using radix-2 DIT – FFT butterfly algorithm.	2	4	4	2	4	6	8	3	2	8	10	4	3	8	6	2	10		
2	4	4	2																	
4	6	8	3																	
2	8	10	4																	
3	8	6	2																	
Q4 (a)	Explain with the help of example region growing technique in image segmentation.	10																		
Q4 (b)	Explain smoothing and sharpening filters with example. OR Explain the following methods for zooming in image enhancement i) Linear interpolation ii) Replication	10																		
Q5 (a)	A 64* 64 image, represented by 8 bits/pixel has following gray level distribution. Perform histogram equalization and show plots of original and equalized image. <table><tr><td>Gray level</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td></tr><tr><td>No of Pixels</td><td>128</td><td>75</td><td>280</td><td>416</td><td>635</td><td>1058</td><td>820</td><td>684</td></tr></table>	Gray level	0	1	2	3	4	5	6	7	No of Pixels	128	75	280	416	635	1058	820	684	10
Gray level	0	1	2	3	4	5	6	7												
No of Pixels	128	75	280	416	635	1058	820	684												
Q5 (b)	Explain Huffman coding and decoding image compression technique. OR Explain with the help of block diagram JPEG lossy compression technique.	10																		

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End Semester Examinations

Nov – Dec 2018

Max. Marks: 100

Duration: 3 Hrs.

Class: TYB.Tech

Semester: VI

Name of the Course: mobile cellular and Ad Hoc Networks

Branch: Computer

Course Code: UCEC604

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Marks
Q 1 (a)	Explain different GSM logical channels.	(10)
Q 1 (b)	Explain CDMA architecture in detail. OR What is FHSS? Explain with it's types, transmitter, receiver, advantages and disadvantages.	(10)
Q2 (a)	Explain GPRS architecture with the help of diagram	(10)
Q2 (b)	Explain W-CDMA frame structure and it's two physical channels. OR Explain VOLTE architecture in detail.	(10)
Q2 (c)	Explain different numbers used in GSM To locate an MS and to address the MS	(5)
Q3 (a)	How synchronization take place in WLAN? Describe the steps for roaming between access points in WLAN.	(10)
Q3 (b)	What is the use of different inter-frame spacing in WLAN? Also explain this different inter-frame spacing in WLAN with diagram. OR What are benefits of reservation schemes? How are collisions avoided during data transmission, why is the probability of collisions lower compared to classical Aloha? What are disadvantages of reservation schemes?	(5)
Q3 (b)	Explain system architecture of 802.11.	(5)
Q4 (a)	Explain any one reactive protocol with example.	(10)
Q4 (b)	Explain DSDV in detail with damping fluctuation. OR Explain ZRP routing Protocol	(10)

Q5 (a)	Describe the overview of IOT hardware platform with respect to device level and network level.	(10)
Q5 (b)	How data dissemination take place in sensor network? OR Differentiate between VANET and MANET.	(5)

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10-12-18(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
Nov – Dec 2018

Max. Marks:20

Class: TY B. Tech

Name of the Course: Mobile Cellular and Ad hoc Network

Course Code: UCEC604 (For specially Abled)

Duration:

Semester: VI

Branch: COMP

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Q. No.		Max. Marks
Q.1	State True or False for following statements: 1. Handoff does not require voice and control channel to be allocated to channels associated with the new base station. 2. In GSM only TDMA is used 3. The cells or subdivisions of a geographical area are always hexagonal. 4. AODV is reactive protocol. 5. The Mobile station can be identified using Subscriber Identity Module (SIM)	5
Q.2	What is the disadvantage of guard channel? a) Efficient utilization of spectrum b) Cross talk c) Near far effect d) Reduce total carried traffic	1
Q.3	Duration of GSM time slot including guard space is _____ μ s.	1
Q.4	_____ Algorithm is used for authentication in GSM.	1
Q.5	In Reverse Path Forwarding (RPF), router receives a packet and extracts the A. Protocol Address B. Source Address C. IP Address D. Standard Address	1
Q.6	Specifications for a wireless LAN are called A. Standard 802.3z. B. Standard 802.3u. C. Project 802.3. D. IEEE 802.11.	1
Q.7	The basic GSM is based on _____ traffic channels. A) connection oriented. B) connection less. C) packet switching. D) circuit switching.	1

Q.8	Two bit type field value in MAC frame for control is a. 00 b. 10 c. 01 d. 11	1
Q.9	_____ is the operating frequency of 802.11b.	1
Q.10	The 2G GSM technology uses a carrier separation of a. 1.25 MHz b. 200 KHz c. 30 KHz d. 300 KHz	1
Q.11	Components of UTRAN are a. UE, CN b. Node B, RNC, UE c. Node B, RNC, UE, CN d. None of the above	1
Q.12	A soft handover is not supported by Core Network in UMTS a. True b. False	1
Q.13	OFDMA stands for _____	1
Q.14	minimum value and maximum value of Contention window size in wireless LAN: a. 7, 255 b. 15, 255 c. 7, 127 d. 15, 127	2
Q.15	_____ routing protocol is used in Wireless Sensor Network	1